CareGuide Rural AI

Rural Healthcare Information & Navigation Assistant

1. Introduction

Access to reliable healthcare information remains a challenge for many people living in rural areas. Finding suitable hospitals, learning about available medical services, and understanding medical reports and finding clear explanations of medical terms can be challenging for people without a medical background.

CareGuide Rural AI is a web-based application designed to make healthcare information easier to access and understand. It will allow users to ask health-related questions in Bangla through text or voice, search for nearby hospitals and healthcare services, and receive simplified explanations of medical documents.

The system will use AI to understand user queries, retrieve relevant information from trusted healthcare sources, and explain medical documents in simple Bangla. By bringing these features together in one platform, the project aims to help rural communities access healthcare information more conveniently. It will provide general information and navigation support, not medical diagnosis or treatment recommendations.

1.1 Background Overview

The healthcare sector is increasingly using digital technologies to improve access to healthcare information and services. In rural Bangladesh, people mainly depend on community clinics, local hospitals, healthcare workers, and telemedicine services for basic healthcare support. However, healthcare information is often scattered across different sources, making it difficult to find the right healthcare facility or understand medical reports.

AI technologies such as **LLMs, RAG, speech recognition, and image understanding** can help make healthcare information easier to access and understand. However, these capabilities are not commonly combined into a single **Bangla-first platform designed for rural users.**

CareGuide Rural AI aims to address this gap by combining AI-based question answering, voice interaction, medical document explanation, and healthcare facility search in one web application. The system will focus on providing accessible, easy-to-understand healthcare information while directing users toward relevant healthcare facilities when needed.

1.2 Problem Statement

People in rural communities may experience difficulties accessing healthcare information because relevant information is often distributed across multiple sources, presented in technical language, or available through interfaces that require users to type and search manually.

The main problems addressed by this project are:

- Difficulty finding hospitals, departments, and diagnostic services based on location and healthcare needs.
- Limited access to healthcare information presented in simple Bangla.
- Difficulty understanding medical reports and other healthcare documents.
- Dependence on manual searching and interpreting information from different sources.
- Limited accessibility of conventional web platforms for users who prefer voice-based interaction.

Existing healthcare websites and directories often require users to search across multiple platforms and interpret information themselves. General-purpose AI systems may not always provide reliable, locally relevant answers. Therefore, a web-based solution is needed to combine healthcare information, hospital search, and medical document explanation in one platform.

CareGuide Rural AI will combine healthcare information, hospital search, and document explanation in one web application. AI will help users find relevant information and understand medical documents in simple Bangla. The project will focus on a selected area and a curated healthcare knowledge base to keep it feasible.

1.3 Objectives

General Objective

To develop a web-based healthcare information and navigation assistant that helps rural users access reliable healthcare information, find healthcare services, and understand medical documents through a simple and user-friendly Bangla interface.

AI Objectives

1. To develop an AI-powered question-answering system that understands healthcare-related queries in Bangla.
2. To retrieve relevant information from a curated healthcare knowledge base and generate informative responses.
3. To provide AI-assisted hospital and healthcare service search based on users' location and requirements.
4. To enable Bangla voice interaction through speech-to-text and text-to-speech technologies.

5. To process uploaded medical documents and explain their readable contents in simple Bangla.
6. To evaluate the **accuracy, relevance, and usability** of the AI-generated responses.

1.4 Scope

In-Scope Features

- Web-based application with a responsive, Bangla-first interface.
- Text-based healthcare question answering.
- Bangla voice input and audio responses.
- Hospital and healthcare service search by location.
- Retrieval of healthcare information from a curated knowledge base.
- AI-assisted selection of appropriate search and information retrieval tools.
- Uploading and explaining supported medical document images.
- Displaying information sources and relevant disclaimers.
- Basic conversation history.
- Administrative management of hospital and healthcare information.

Out-of-Scope Features

- Automated disease diagnosis.
- Prescription generation or modification.
- Medication dosage recommendations.
- Emergency diagnosis or automated medical triage.
- Direct integration with hospital electronic health record systems.
- Online appointment booking and payment processing.
- Nationwide hospital coverage in the initial version.
- Guaranteed interpretation of handwritten prescriptions or medical documents.
- Training a large language model from scratch.

AI Boundaries

The AI components will understand user queries, retrieve relevant information, process supported documents, and generate simple Bangla explanations. The system will **not diagnose diseases, make clinical decisions, prescribe medications, or replace qualified healthcare professionals**.

The system will indicate when information is insufficient, or a document cannot be read reliably. Healthcare responses will be supported by available sources, while hospital and service information will be based on the records maintained in the system.

1.5 Stakeholders

Stakeholder	Role / Involvement
Rural Residents	The primary users who will use the website to get healthcare information in Bangla and find nearby hospitals and healthcare services.
Healthcare Professionals	They provide professional healthcare services. The system may help users understand general health information, but it does not replace medical advice.
Hospitals and Diagnostic Centers	They are the sources of information about hospital locations, departments, and available healthcare services.
System Administrator	The administrator manages hospital information, updates healthcare resources, and maintains the system.
Development Team	The team designs, develops, tests, and maintains the web application and its AI-based features.

1.6 Proposed Solution

CareGuide Rural AI will be developed as a user-friendly web application that provides healthcare information in Bangla through text, voice, and document uploads.

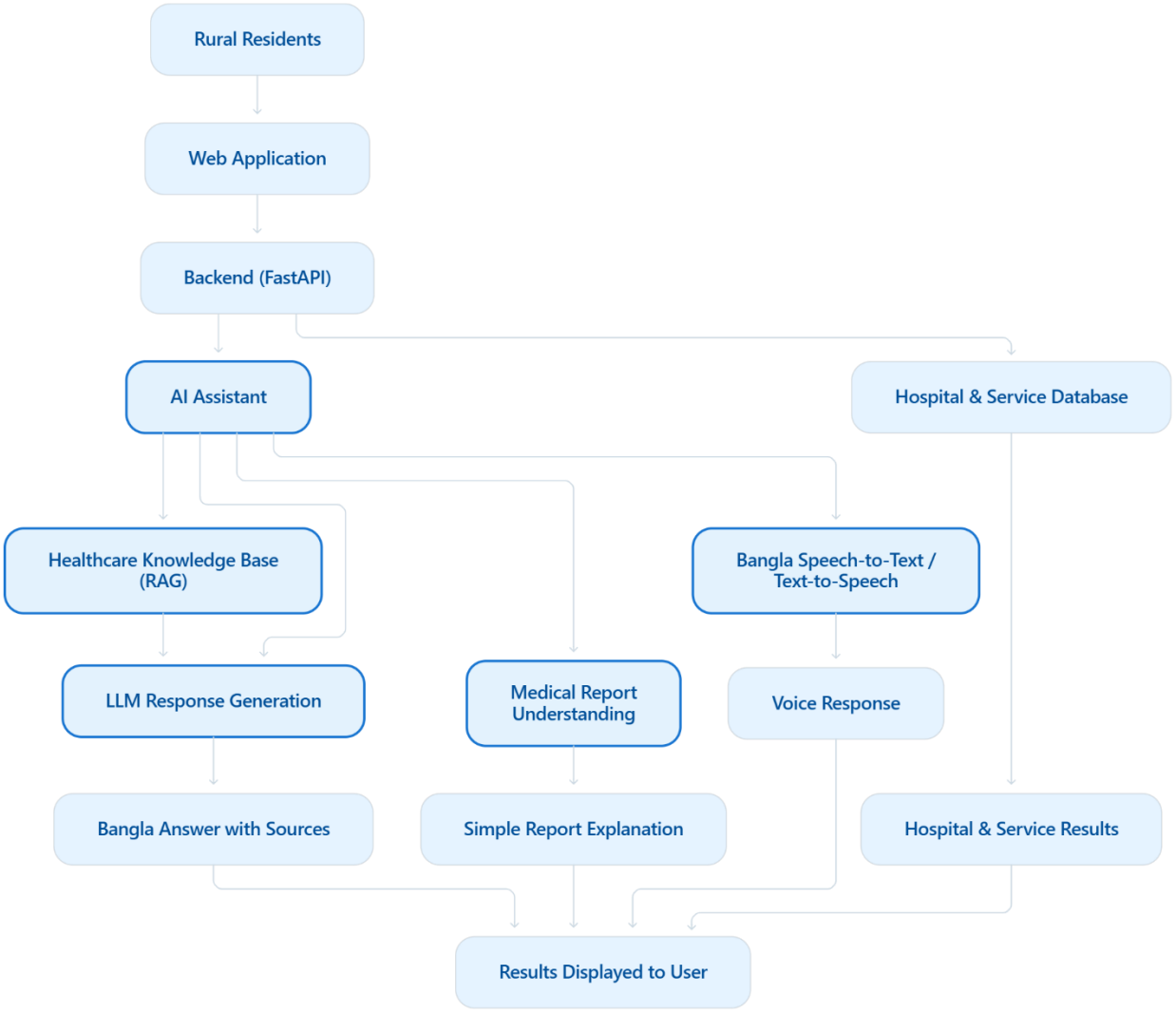
Users will be able to ask health-related questions, find nearby hospitals and diagnostic services, and upload medical reports for simple explanations. The AI assistant will use trusted healthcare resources to generate clear answers, while the hospital finder will provide relevant location and service information. Voice interaction will allow users to ask questions and listen to responses.

Key Features

- **AI Healthcare Assistant:** Answers general healthcare questions in simple Bangla using trusted information.
- **Hospital and Service Finder:** Helps users find nearby hospitals, departments, and diagnostic services.
- **Medical Report Explainer:** Uses AI to read supported medical document images and explain their contents in simple language.
- **Bangla Voice Interaction:** Converts spoken questions into text and generates audio responses.
- **Source-Based Answers:** Shows supporting sources for healthcare information where available.

System Workflow

Users interact with the web application through text, voice, or document uploads. The backend processes each request and sends it to the relevant component. AI handles question understanding, healthcare information retrieval, response generation, and document explanation. Hospital searches use the hospital database. Finally, the system displays the result or provides it as a Bangla audio response.



1.7 AI Features

Existing AI Features

Several existing AI systems provide features relevant to CareGuide Rural AI. However, they have limitations when used for rural healthcare information and navigation.

Existing System	Available AI Features	Limitations
Microsoft Copilot Health	Health Q&A, health information retrieval, lab-result understanding, health insights, and care navigation	Currently available as a limited US preview. It is not specifically designed for rural users or Bangla-first healthcare support.
ChatGPT	Natural-language Q&A, voice interaction, image and document understanding	It is a general-purpose AI system rather than a rural healthcare platform, and responses may require verification for important medical information.
Google Gemini	Text and image understanding, multimodal interaction, and response generation	It is a general-purpose AI system and does not specifically provide rural healthcare navigation or a dedicated local healthcare knowledge base.

Proposed AI Features

CareGuide Rural AI will combine these AI capabilities to provide healthcare information and navigation through a single web application.

AI Feature	Input	Output	Manual Work Replaced
Bangla Healthcare Assistant	User's health-related question	Simple Bangla answer with supporting sources	Searching multiple websites and simplifying information manually
Medical Report Explainer	Uploaded medical report image	Extracted information and a simple explanation in Bangla	Manually reading and understanding report terminology
Bangla Voice Interaction	Spoken question	Text transcription and spoken Bangla answer	Typing questions and reading responses
AI-Based Information Retrieval	User's question	Relevant information from the trusted healthcare knowledge base	Manually searching through healthcare documents

AI Techniques

Technique	Purpose
Large Language Model (LLM)	Understands questions and generates Bangla responses.
Retrieval-Augmented Generation (RAG)	Retrieves relevant information from trusted healthcare documents before generating answers.
AI Agent / Tool Calling	Selects suitable tools, such as healthcare information retrieval or hospital search.
Computer Vision / OCR	Extracts information from uploaded medical report images.
Speech-to-Text (STT)	Converts spoken Bangla questions into text.
Text-to-Speech (TTS)	Converts generated answers into Bangla audio.
Semantic Search	Finds relevant healthcare information based on the meaning of a question.

AI is the core of CareGuide Rural AI because its main purpose is to help users understand healthcare information in simple Bangla through natural-language questions, voice interaction, and medical document explanations. Without AI, the system would be limited to displaying stored information and hospital listings. It would not be able to understand users' questions, explain medical reports, or generate personalized, easy-to-understand responses. Therefore, AI is essential to achieving the project's main objective.

2. System Requirements

Requirements are written as testable statements. FR = functional requirement, AIR = AI requirement, NFR = non-functional requirement

2.1 Functional Requirements

FR	FR-n	Stakeholder	Description
User Registration	FR-1	All Users	Register new users using valid identity and account information.
Secure Login	FR-2	All Users	Allow registered users to securely access their accounts.

User Query	FR-3	All Users	Ask healthcare-related questions using Bangla, English, or mixed text.
Voice Query	FR-4	Rural Users	Record and submit healthcare-related questions using voice input.
Document Upload	FR-5	All Users	Upload prescriptions, test reports, referrals, and other healthcare documents.
Text and Audio Response	FR-6	All Users	View system responses as text and listen to them as audio.
Knowledge Base Retrieval	FR-7	All Users	Retrieve relevant healthcare information and display the source when available.
Healthcare Navigation	FR-8	Rural Users	Find relevant services with location, timing, and preparation details.
Conversation History	FR-9	All Users	Maintain session history for understanding follow-up questions.
Admin Login	FR-10	Admin	Provide secure admin access for managing healthcare information.
Knowledge Base Management	FR-11	Admin	Upload, update, delete, and re-index healthcare documents and FAQs.
Service Data Management	FR-12	Admin	Manage healthcare services, departments, locations, and timings.
Profile Management	FR-13	All Users	Allow users to view and update their basic profile information.
Document Explanation	FR-14	All Users	Explain uploaded healthcare documents and reports in simple Bangla.
Service Search	FR-15	All Users	Search for relevant healthcare services using natural-language queries.
User Feedback	FR-16	All Users	Allow users to provide feedback on system responses.
System Monitoring	FR-17	Admin	Log queries, tool calls, and system errors for monitoring and evaluation.

2.3 Non-Functional Requirements

FN	NFR-n	Stakeholder	Description
Response Time	NFR-01	All Users	The system shall provide a response within 5 seconds for normal text-based queries.
Usability	NFR-02	All Users	The system shall provide a simple and user-friendly interface for users with limited technical literacy.
Language Support	NFR-03	All Users	The system shall support Bangla as the primary language and handle English and mixed-language queries.
Data Security	NFR-04	All Users	The system shall protect user documents and personal information from unauthorized access.
Authentication Security	NFR-05	Admin	The system shall use secure authentication and authorization for administrator access.
Reliability	NFR-06	All Users	The system shall maintain reliable retrieval from the verified healthcare knowledge base.
Availability	NFR-07	All Users	The system shall remain available under normal expected user load.
Maintainability	NFR-08	Admin	The system shall use a modular design for easy updates to AI models, knowledge sources, and services.
Monitoring	NFR-09	Admin	The system shall log relevant errors and AI operations while minimizing unnecessary personal data storage.
Scalability	NFR-10	Admin	The system shall support additional users, healthcare information, and services in future versions.

2.3 AI Requirements

AI Feature	AI-n	Stakeholder	Description
LLM Query Understanding	AI-01	All Users	Understand user queries and generate natural-language responses.
RAG Retrieval	AI-02	All Users	Retrieve relevant information from the verified healthcare knowledge base.
Semantic Search	AI-03	All Users	Find relevant information based on the meaning of user queries.
AI Agent	AI-04	All Users	Perform multi-step retrieval and tool-based tasks when required.

Speech AI	AI-05	Rural Users	Process Bangla voice input and generate Bangla voice responses.
Multimodal AI	AI-06	All Users	Process uploaded healthcare images and documents.
Grounded Response	AI-07	All Users	Generate responses based on retrieved or available information.
Insufficient Information Handling	AI-08	All Users	Indicate when sufficient information is unavailable.

3. Tools and Technologies

Technology	**Main Purpose**
JavaScript + React.js	Frontend web application development
Tailwind CSS	Responsive UI design
Python + FastAPI	Backend API and application logic
Pydantic + Uvicorn	Backend data validation and server execution
PostgreSQL + pgvector	Structured data storage and vector search for RAG
Google Gemini API	LLM-based healthcare question understanding and response generation
Gemini Multimodal Model	Medical document/image understanding
LangChain	RAG pipeline and LLM integration
LangGraph	AI agent and tool-calling workflow
BGE-M3	Multilingual text embeddings
Google Cloud Speech-to-Text	Bangla voice-to-text conversion
Google Cloud Text-to-Speech	Bangla text-to-voice conversion

4. Project Timeline and Work Plan

The proposed development period is 12 weeks. The timeline assumes a three-member team and may be adjusted according to the course schedule and available resources.

Week	Milestone	Tasks	Expected Deliverable
Week 1	M1: Requirement Analysis	Finalize scope, use cases, target users, and system requirements	Approved requirements
Week 2	M2: System Design	Design architecture, database schema, API, AI workflow, and UI	Design documents
Weeks 3–4	M3: Backend and Database	Develop FastAPI backend, PostgreSQL database, authentication, and service management	Working backend
Weeks 5–6	M4: RAG Pipeline	Collect healthcare documents, create embeddings, and implement retrieval	Working RAG system
Week 7	M5: AI Agent Integration	Implement intent routing, tool calling, and agent workflow	Integrated AI agent
Week 8	M6: Multimodal Processing	Implement image/PDF processing and healthcare document explanation	Document explainer
Week 9	M7: Voice Integration	Integrate Bangla speech-to-text and text-to-speech	Voice-enabled assistant
Week 10	M8: Frontend Integration	Connect frontend with backend and AI features and improve UI	Integrated application
Week 11	M9: Testing and Evaluation	Test AI responses, retrieval, voice, documents, safety, and usability	Evaluation report
Week 12	M10: Deployment and Submission	Deploy the system and complete documentation, presentation, and demo	Final submission

5. Optional Sections

5.1 Existing System

Several existing AI systems provide healthcare-related information and multimodal interaction. **Microsoft Copilot Health** provides health information, health-related question answering, and care navigation. **ChatGPT** supports natural-language interaction, voice conversations, and image/document understanding. **Google Gemini** provides multimodal AI capabilities for understanding text and images.

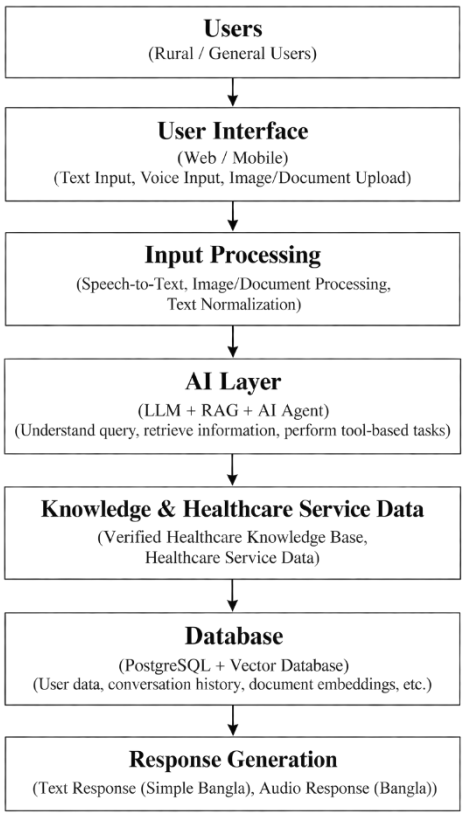
However, these systems are not specifically designed as a **Bangla-first rural healthcare information and navigation platform**. They do not provide the complete combination of localized healthcare information, hospital/service navigation, RAG-based trusted knowledge retrieval, medical document explanation, and Bangla voice interaction proposed in CareGuide Rural AI.

5.2 System Architecture

The proposed system will follow a layered architecture consisting of:

- **User Interface:** Text, voice, and image/document input.
- **Input Processing:** Speech-to-Text and document/image processing.
- **AI Layer:** LLM, RAG, semantic search, and AI Agent.
- **Knowledge & Tool Layer:** Verified healthcare knowledge base and healthcare service data.
- **Database Layer:** PostgreSQL and vector database.
- **Response Layer:** Simple Bangla text and voice responses.

Main flow:



5.3 Ethical, Legal, and Social Considerations

- **Data Privacy:** Personal information and uploaded medical documents will be handled securely and will not be unnecessarily shared.
- **AI Bias:** Reliable and curated healthcare sources will be used to reduce inaccurate or biased information.
- **Medical Safety:** The system will provide general healthcare information only and will not diagnose diseases or prescribe medicines.
- **Human Oversight:** Users will be advised to consult qualified healthcare professionals for diagnosis, treatment, and emergencies.
- **Accessibility:** Simple Bangla language and voice interaction will make the system easier to use for rural users with limited digital literacy.

5.4 Future Work

Future versions of CareGuide Rural AI can include:

- Expansion of the hospital and healthcare service database to cover more regions of Bangladesh.
- Integration with verified hospital appointment and scheduling systems.
- Support for additional regional languages and dialects.
- Improved voice interaction for users with limited literacy.
- Integration with verified health records, where legally and technically appropriate.
- Personalized healthcare information based on user preferences and history, with proper privacy controls.
- Mobile application development for Android and iOS.
- Continuous evaluation and improvement of AI responses using expert-reviewed healthcare datasets.